The Fate of Uncertainty

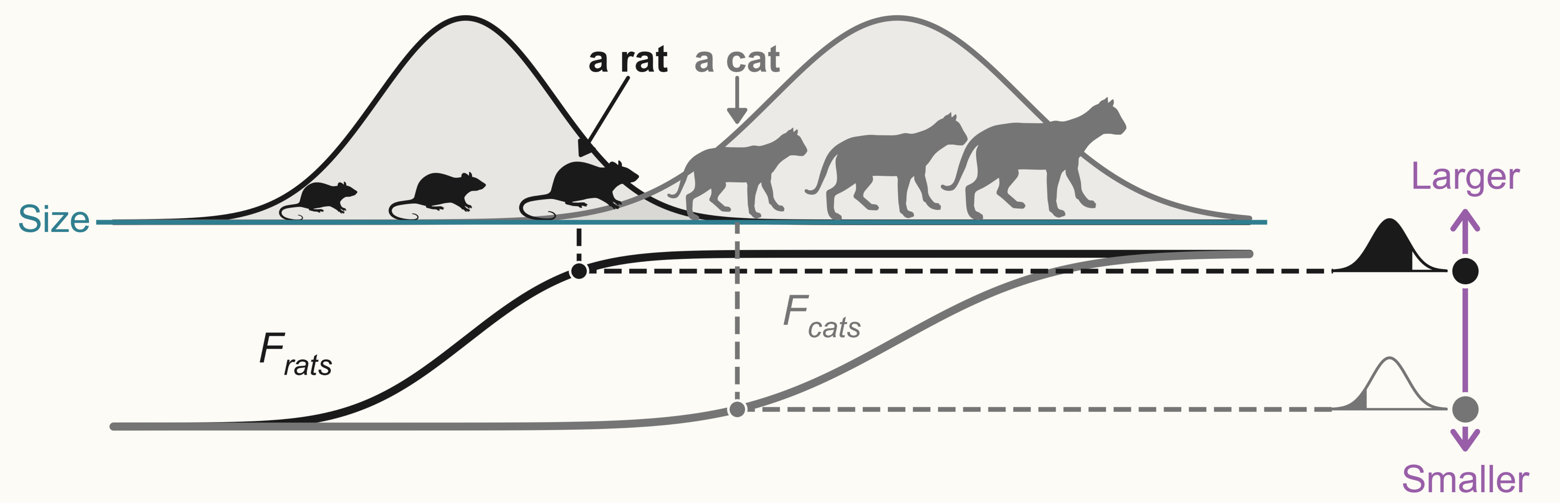
in the Transformation from Absolute to Relative Magnitude

Joonoh Park[1], Sang-Hun Lee[1†]

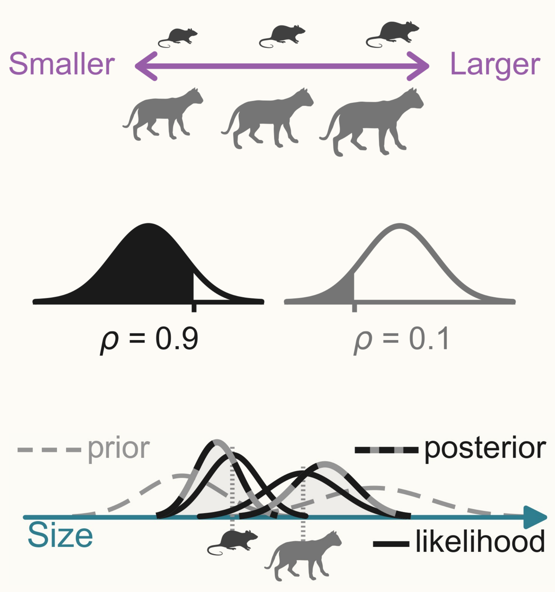
[1] Department of Brain and Cognitive Sciences, Seoul National University

1 · Background

Transformation from Absolute to Relative Magnitude



- Human judgments are context-dependent: the same physical magnitude is judged differently depending on where it falls within the current context[1,2].
- Such judgments transform an absolute magnitude (θ) into its percentile (ρ) within a context (c): $\rho = F_c(\theta)$, where F_c is the context's cumulative distribution function[3].
- A Bayesian observer maintains a posterior probability distribution over candidate states to minimize expected loss[4–6].

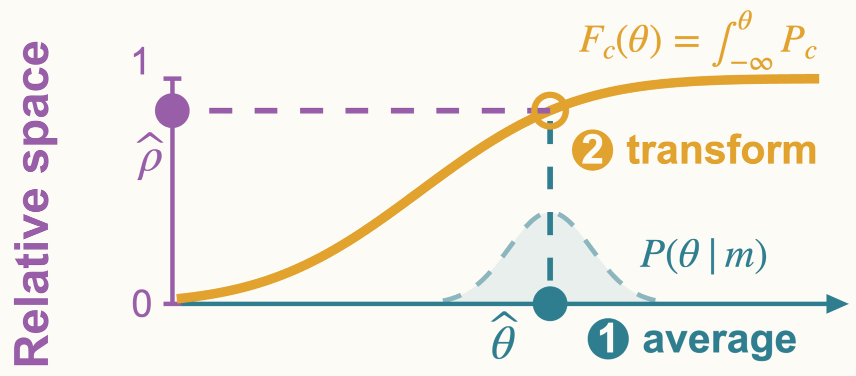


What happens to this uncertain belief when an absolute magnitude is transformed into a relative coordinate?

2 · Competing Hypotheses

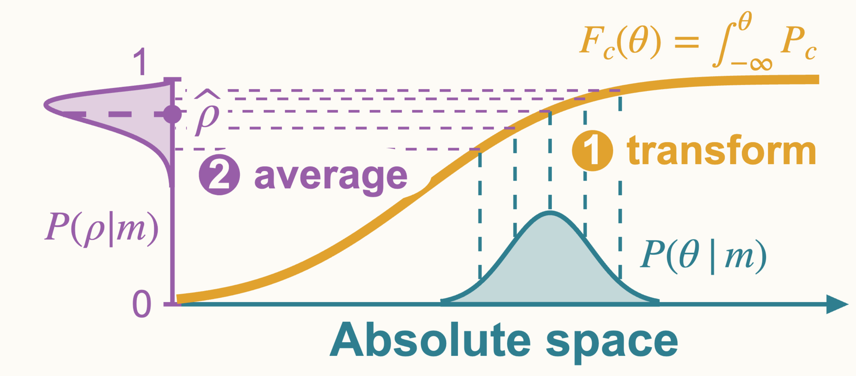
Point-estimate hypothesis (H_1)

Average, then transform. Only the posterior mean (point estimate) is transformed into relative space.



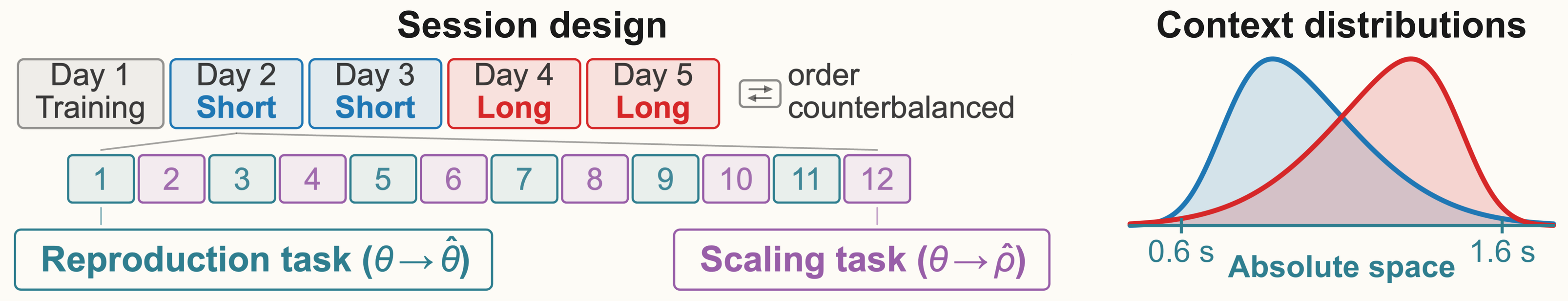
Distributional hypothesis (H_2)

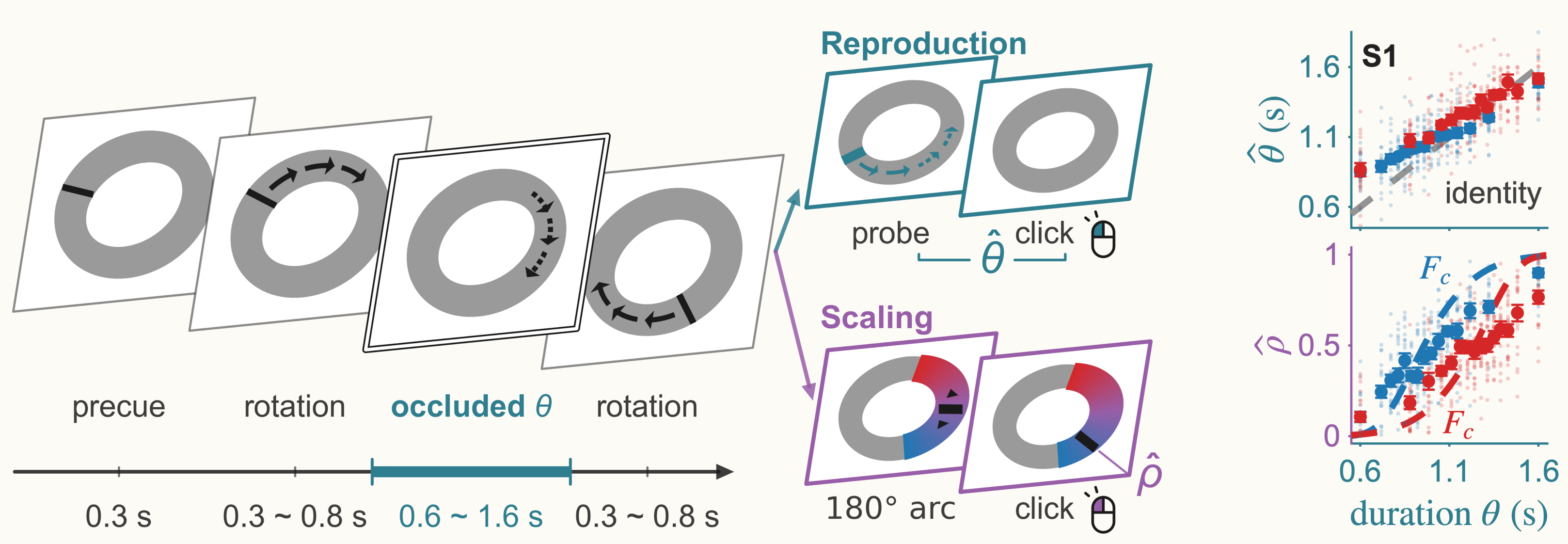
Transform, then average. The full posterior distribution is transformed into relative space, where its expected value is computed.



3 · Experimental Paradigm

- Stimulus durations (θ) are sampled from 15 uniformly spaced relative percentiles in either a right-skewed (**Short**) or left-skewed (**Long**) context.
- A rotating bar is occluded for a duration θ.
- **Reproduction trial**: a probe bar starts rotating at a random position and speed, then vanishes; observers click to match the elapsed duration.
- **Scaling trial**: observers indicate the relative percentile of θ within the context by clicking along an arc.





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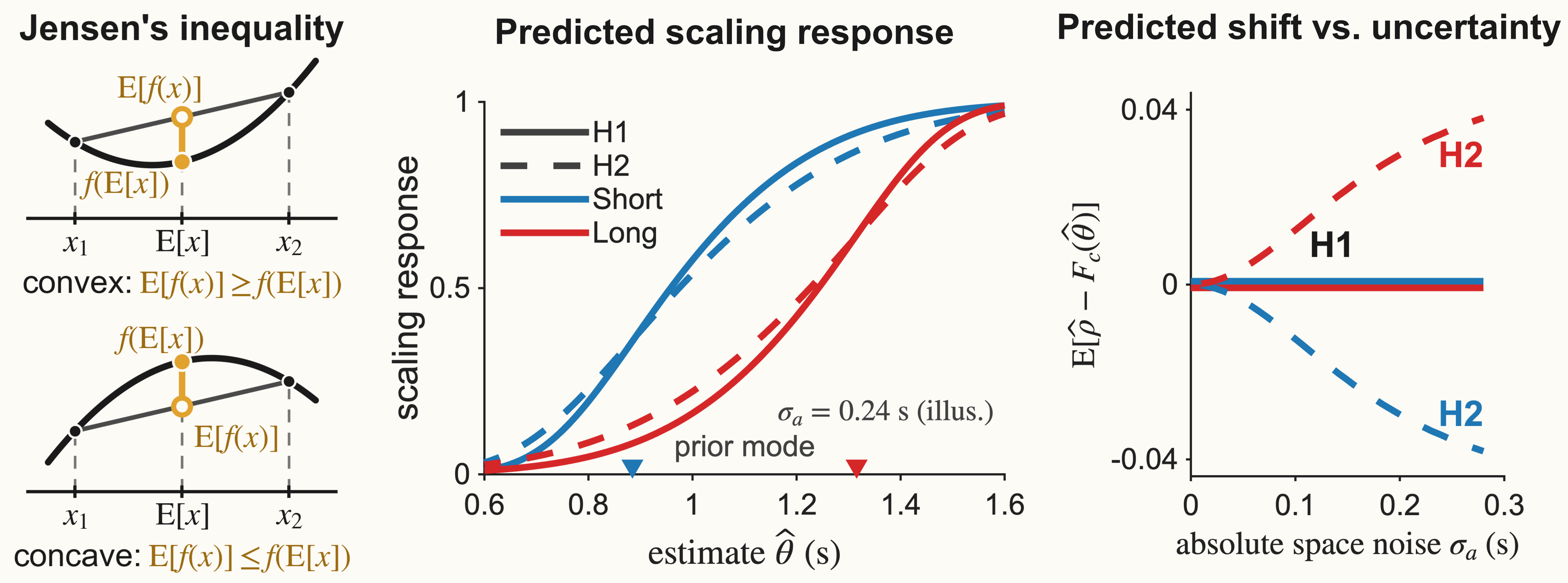
4 · Predictions

The distributional hypothesis (H_2) predicts a systematic shift of the scaling response away from the veridical CDF; its sign follows the CDF's local curvature and its size grows with uncertainty.



5 · Results

Fig 1 · Scaling response (Subject #6)

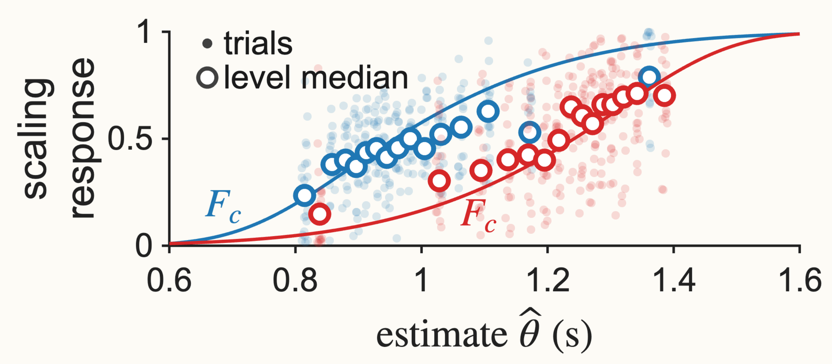


Fig 1 · Scaling response. Absolute estimates were derived from reproduction trials via Bayesian model fitting and substituted into the same subject's scaling trials. A single subject is shown.

Fig 2 · Distribution-dependent shift

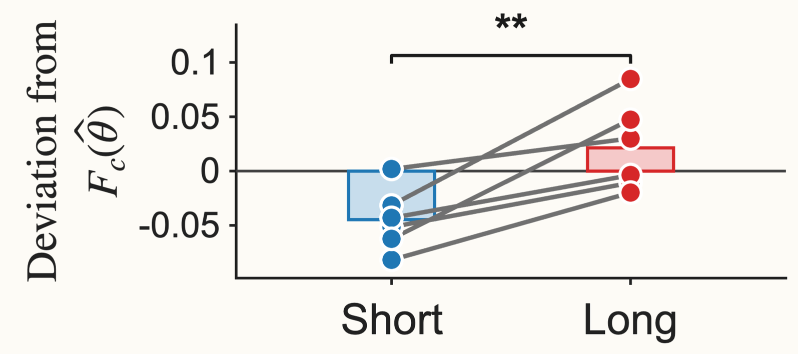


Fig 2 · Distribution-dependent shift. Median response deviation from its CDF, averaged over estimates in [0.88, 1.32] s per participant. The difference between distributions was reliably above zero (paired $t(5) = 4.25$, $p = 0.008$).

Fig 3 · Model architecture

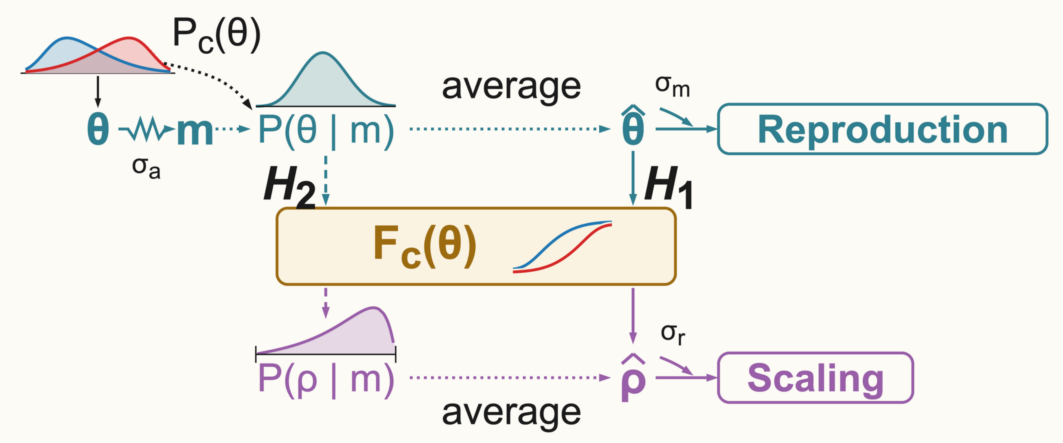


Fig 3 · Model architecture. A noisy measurement is integrated with the contextual prior to form a belief distribution. Reproduction reports the belief mean under motor noise; scaling transforms the belief according to each candidate rule under response noise.

Fig 4 · Predictive performance

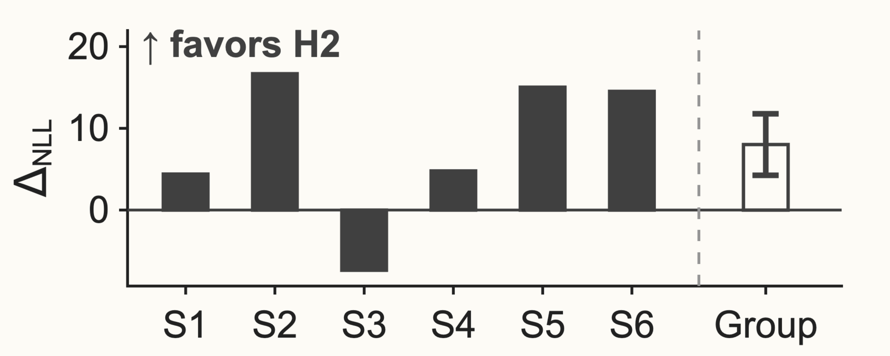


Fig 4 · Predictive performance. Difference in negative log-likelihood (Δ_{NLL}) between H_1 and H_2 on scaling responses (5-fold cross-validation). Positive values favor the distributional hypothesis (H_2). 5 out of 6 participants showed higher predictive performance under H_2.

Fig 5 · Generative performance (Subject #6)

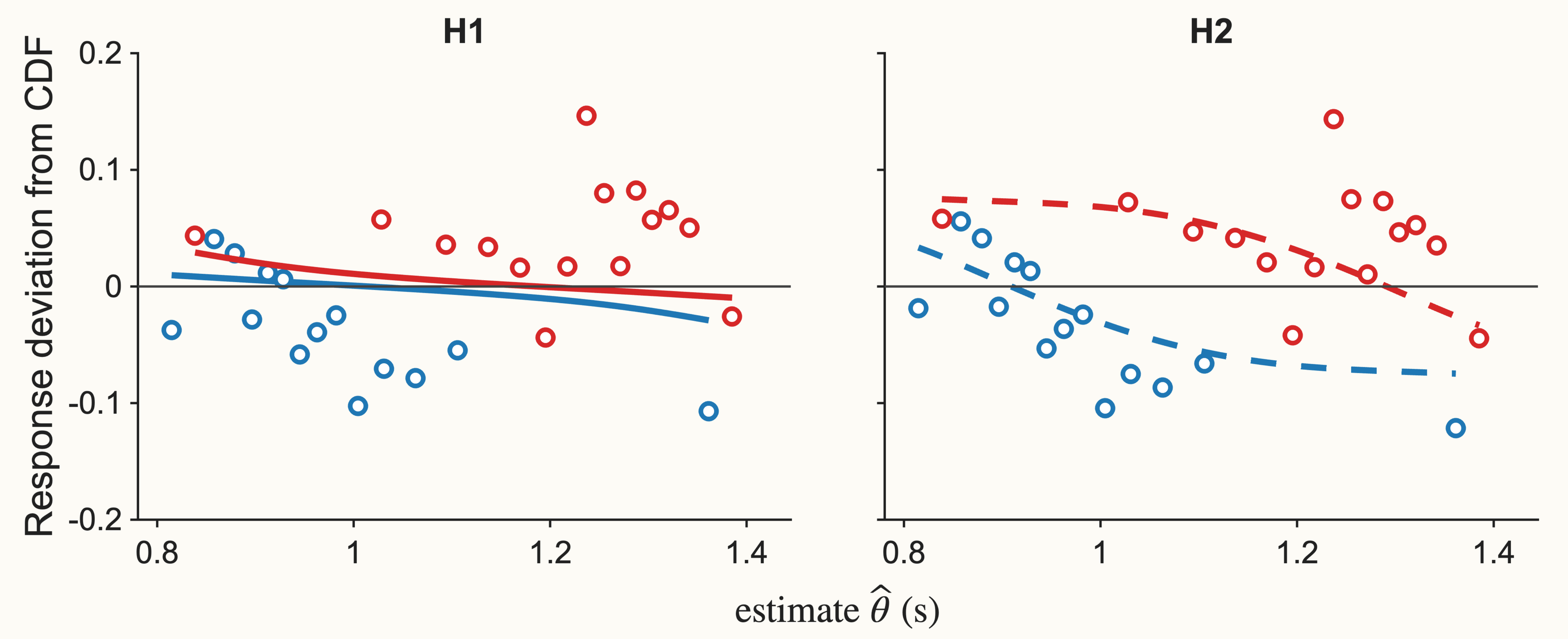


Fig 5 · Generative performance. Observed data (open circles) and model simulations (lines) are plotted against estimated durations. The vertical axis represents the response deviation from the veridical CDF. Lines depict median responses generated by each hypothesis (H_1 and H_2) using individual best-fit parameters, whereas open circles denote observed level medians. A single subject is shown.

6 · Conclusion & Further work

- **Main finding:** Empirical data exhibit the systematic, curvature-dependent shift predicted by the distributional hypothesis.
- **Trial-wise uncertainty:** Posterior beliefs were inferred rather than directly measured. Follow-up studies will use confidence reports or a wagering paradigm to directly access trial-level belief distributions[7].
- **Single noise level & veridical prior:** The current task tested a single noise level under the assumption of a veridical prior; future work will manipulate noise levels to recover subjective priors[8].